

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Hydrotest

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Revision History

Rev No.	Date	Description of Revision
R01	10-07-22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-009

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

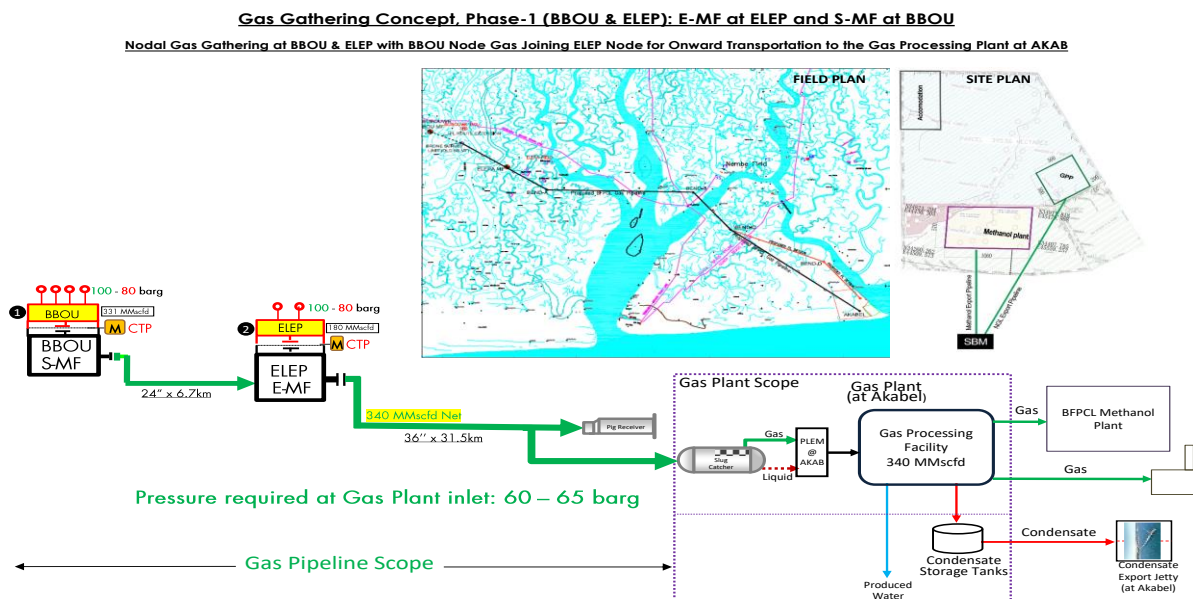


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification defines the pressure and leak test requirement and acceptance criteria for piping systems. This specification is restricted to rigid metallic piping system constructed from carbon steel, clad steel, stainless steel and the like. It does not address flexible pipes or cement-lined pipelines, or pipelines constructed from other materials such as GRP or RTRP.

This Specification is applicable to projects or works associated with the GAS PROCESSING AND METHANOL COMPLEX PROJECT for Brass Fertilizer and Petrochemical Company Limited (BFPCL).

The Specification prescribes the requirements for pressure testing of pipe spools and installed piping systems under the jurisdiction of the of ASME B31.3 Code.

The purpose of this Specification is to prescribe the methods, tasks, scope, and criteria for testing pipe spools and installed piping systems for GAS PROCESSING AND METHANOL COMPLEX PROJECT

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Scope

The Specification defines the minimum requirements for pressure testing of pipe spools and installed piping systems in line with ASME B31.3.

The following are excluded from the requirements of this Specification:

- a. Any package unit previously tested by the Manufacturer in accordance with the applicable codes.
- b. Atmospheric sewers and drains.
- c. Plumbing systems, which are tested in accordance with the applicable plumbing codes.
- d. Lines and systems open to the atmosphere provided it is Category "D" fluid.
- e. Instrument piping lead lines between the root valve at the process or utility line and the connected instrument and control piping used to connect air or hydraulically operated control apparatus.
- f. Boiler external piping under the jurisdiction of ASME B31.1.
- g. Liquid petroleum transportation pipelines systems under the jurisdiction of ASME B31.4.
- h. Gas transmission and distribution pipelines systems under the jurisdiction of ASME B31.8.
- i. No deviation shall be made from this Specification without written authorization from Brass Fertilizer and Petrochemical Company Limited (BFPCL).

2. Regulations, Codes And Standards

Document Number	Document Title
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.21	Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.34	Valves - Flanged, Threaded, And Welded End
ASME B31.3	Process Piping

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1 Abbreviations

Abbreviation	Meaning
Company	Brass Fertilizer and Petrochemical Company Limited (BFPCL).
Contractor	The Company awarded the contract to perform the work described herein.
Project	GAS PROCESSING AND METHANOL COMPLEX PROJECT
Calculated Test Pressure	The test pressure determined in accordance with Section 7.3 of this Specification.
Designated Engineer	Qualified Engineer designated by company and contractor for the purpose of carrying out hydrotesting.
LVP	Low Vapour Pressure (LVP)
Maximum Test Pressure	The highest allowable test pressure gage reading. (The pressure test rating of the weakest component in the test system)
Minimum Design Metal Temperature (MDMT)	Design Minimum Temperature for which the material is normally suitable without impact testing other than that required by the governing material specification.
Minimum Test Pressure	The lowest allowable test pressure gage reading. (The calculated test pressure excluding the additional pressure resulting from the static head of the test fluid.)
System Test	A test that includes multiple sections of piping, having the same or different design pressures, but which are tested together at one time using a single test procedure and pressure.
Test Temperature	The minimum temperature occurring during the test period of the metal temperature of the piping being tested (including any vessels or equipment included in the test).

3. GENERAL

3.1. Pressure Testing

All pressure testing shall be in accordance with pressure piping codes ASME B31.3 or Company specifications.

The test pressure for each line shall be specified in the piping line list or piping Isometric drawings.

3.2. Closure Weld

Field closure welds shall be examined in accordance with ASME B31.3 para. 345.2.3 (c) in lieu of pressure testing. A hydrotest waiver form shall be completed for all cases. For registered piping systems, examination of closure welds in lieu of pressure testing requires approval of the local jurisdiction, in accordance with Company requirements.

4. PRODUCTS

4.1. Pressure Test Blinds

Special length bolting and test gaskets may be required during testing, and shall be removed after test completion and replaced with standard line class bolts and new service gaskets.

A field procedure must be established and care taken to ensure the installation and removal of material which is specified for testing only. Plate material, extra length bolts, and gaskets for testing shall be furnished by the field construction contractor unless dictated otherwise by contract requirements.

5. EXECUTION

5.1. Preparation

5.1.1. General

Subsequent to repairs and heat treatment and prior to initial operation, all installed piping shall be pressure tested except where otherwise qualified in this Specification.

The company inspector shall be notified at least 48 hours in advance of pressure testing.

The test shall be hydrostatic using water unless there is a possibility of damage due to freezing; or, if the operating fluid or piping material would be adversely affected by water, any other suitable liquid or additives may be used on approval by the COMPANY Engineer. Testing of ferritic piping, however, is limited to the temperature restrictions in this Specification.

If a flammable liquid is used, its flash point shall be no less than 49°C (120°F) and consideration shall be given to the test environment. If testing with glycol-water mixture to prevent freezing, the safety, leak and disposal problem of the test fluid shall be considered.

5.1.2. Vessels, Equipment and Piping Components

The following equipment shall be excluded from all piping pressure tests. The equipment shall be isolated from the system:

- Vessels and tanks
- Heat exchangers
- Rotating machinery, such as pumps, compressors, and turbines
- Safety valves rupture discs, flame arresters, and steam traps.
- Pressure vessels with sophisticated internals.
- Equipment and piping lined with refractoriness.
- Equipment and Supplier furnished piping specifically recommended by the Manufacturer not to be tested
- Expansion joints
- Filters.
- Heat exchangers of which tube sheets and internals have been designed for differential pressure between tube side and shell side.
- In-line process valves (where feasible)

The following equipment and components shall not be included in the system.

The equipment shall be isolated from the system.

5.1.3. Instruments

All in-line Instruments shall be either removed or blocked prior to hydrotesting to prevent damage

during hydrotesting (e.g., control valves, pressure gages, level gages, and flowmeters).

Instrument take-off piping and sampling system piping, up to the first block valve, shall be tested with the piping to which it is connected.

Instrument lead lines, between the first block valve and the instruments to which they are connected, shall be pressure tested to the test pressure of the associated piping. Testing may be performed separately or at the same time as the piping is tested, but the instruments shall be disconnected.

5.2. Test Medium

The test medium shall be water. Clean water that will not corrode and/or damage the test system shall be used for hydrostatic tests. Water containing silt or suspended material shall not be used, and a suitable filter with 40 mesh should be provided in the filling line. COMPANY Designated Engineer shall approve the proposed source of test water.

If water source is not potable, COMPANY Designated Engineer shall consider adding inhibitor to reduce risk of pitting or introduction of microbiologically induced corrosion (MIC) into the piping spools. COMPANY Designated Engineer should also consider biodegradable inhibitor to allow fluid re-entry into the environment.

A mixture of glycol-water shall be used where the ambient temperature may reach less than 0°C (32°F) during testing or prior to dry-out.

5.3. Test Methods and Pressures

- a. Where piping systems of different design pressures are fully welded together, then "cascade" pressure testing will be required, (i.e., complete the pressure test on the higher-pressure system prior to system closure welding and then repeat full Pressure Test Procedure for the test of the combined systems at the lowest pressure).

As large a piping system as practicable including its associated equipment can be hydro tested together as a unit where applicable. The test pressure in the system shall be equal to the test pressure applied to the system component of which the design pressure is the lowest among the system.

- b. Pressure vessels may be included in a test system provided the system test pressure does not exceed the shop test pressure of any of the included items.

5.3.1. Hydrostatic Testing of Piping Designed for Internal Pressure (ASME B31.3)

Hydrostatic test pressure shall be determined per the applicable code, except as follows:

Underground pressure piping in water service and employing non-welded girth joints (such as bell and spigot) shall be hydrostatically tested at 1.5 times the system design pressure. The test pressure shall be maintained for 2 hours while the leakage rate is measured and the joints are inspected for leakage.

- The minimum hydrostatic test pressure at any point in the system shall not be less than 1-1/2 times the design pressure.

5.3.2. Commodity Test for Category 'D' Fluid Service only (ASME B31.3)

A piping system used only for Category D Fluid Service, not previously hydrostatic tested in accordance with this Specification, may be tested at the normal operation conditions of the system during or prior to initial operation by examining for leaks at every joint.

5.4. Test Preparation – Field Procedure

Joints, including welds, shall be left uninsulated and exposed for examination during the test, except that joint previously tested in accordance with this Specification may be insulated or covered. All joints that require a leak test shall not be primed and painted prior to leak testing.

Underground portions of piping systems may be tested and covered before testing Above ground portions.

Piping designed for vapour or gas shall be provided with additional temporary supports, if necessary to support the weight of the test liquid.

Lines that are counterweight supported shall be temporarily blocked during testing in order to sustain the weight of the test fluid. Spring hangers that have been provided with stops for carrying the test load normally do not require additional temporary supports; if this is not the case, temporary support must be provided before filling the system.

Before testing, the following procedures shall be carried out:

- Verify that any required heat treatment has been performed and that all NDE requirements are complete.
- Piping systems shall have been completely checked for completeness.
- Piping systems shall be checked to ensure that the entire system can be completely drained after testing.
- Vents or other high point connections shall be opened to eliminate air from lines that are to receive a hydrostatic test.
- System shall be purged of air before hydrostatic test pressure is applied.
- Short pieces of piping that must be removed to permit installation of a blind or blank shall be tested separately.
- Flanged joints at which a blank is inserted to isolate other equipment during a test need not be tested after blank is removed.
- Lines containing check valves shall have the source of pressure located in the piping upstream of the check valve so that the pressure is applied under the seat. If this is not possible, remove or jack up the check valve closure mechanism or remove check valve completely, and provide necessary filler piece or blinds. Removed internals shall be bagged, and placed

close to the valve they were removed from in order that inspectors may verify that the internals have been removed for the test.

- When conducting tests at freezing temperatures, the test shall not take more than 4 hours, and special precautions, such as using glycol/water mixture, shall be observed to avoid freezing damage. Follow precautions to minimize the risk of brittle fracture as noted in this Specification.
- Systems that include expansion joints, which have been removed for pressure testing, shall be investigated to see that any required temporary restraints, anchors, or guides are installed or removed prior to test.
- When a pressure test is required to be maintained for a period of time during which the testing medium in the system would be subject to thermal expansion, provision shall be made for relief of any pressure greater than the maximum test pressure.
- With prior authorization from COMPANY Designated Engineer, piping or sections of piping to be tested may be isolated by closed valves provided the valve closure is suitable for the test pressure. Refer to applicable valve testing standards.
- Drain and vent valves to be in open position and plugged during test to permit testing of valve packing.
- Bypass valves on gate valves provided with centre cavity pressure equalizing bypass shall be open during the test.

5.5. Hydrostatic Test Procedure

Piping systems shall be pressurized slowly and evenly to prevent vibrations. When the pressure has been raised to 100% of the test pressure, a leak check shall be done to check all valves and fittings. If leaks are found, system must be depressurized prior to fixing leaks.

In order to hydrostatic test as much piping as possible at one time, a systems test may be employed.

The minimum test pressure for a system test shall be such that each line in the system is subjected to a test pressure in accordance with Section 5.3 of this Specification.

The maximum system test pressure shall not exceed the pressure test rating of any piping component.

Systems or sections of systems to be tested may be isolated by closed valves, provided the valve body and seat are suitable for the test pressure.

Where a suitable valve is not available, vessels, equipment, or other piping not included in the system pressure test shall be either disconnected from the system or isolated by blinds or other means during the test.

The normal location for the pressure test gauge is at grade near the pressure test pump. Readings may be made at higher points providing the gauge pressure reading and the static head for water between grade and the point of measurement do not exceed the maximum test pressure. Gauges

shall be tagged with the date last calibrated, and this activity shall be recorded.

Hydrostatic test pressure shall be maintained for a sufficient length of time to visually determine whether there are any leaks, but no less than 10 minutes.

Gauges shall be a minimum 100 mm (4") diameter face and shall be graduated over a range of approximately double the intended test pressure but in no case shall the range be less than 1.5 times nor greater than 4 times that pressure.

At least two gauges shall be used per test. One gauge shall be located on the test head and one on the system being tested. Gauge must read within 5% of each other.

The use of NDE in lieu of the hydrotest for the final closure weld requires a hydrotest waiver as described in Section 3.2.

Test procedures and test schedules of lines or systems shall be reviewed by Company prior to the start of the test. Procedures shall include the following:

- Source of test fluid
- Disposition of test fluid
- Provisions to prevent the creation of significant negative pressure (vacuum) during draining of the test fluid, etc
- The specific lines to be included within a system test
- Provisions for temporary supports, vent, drain, fill connections, and gage locations.

All tests shall be witnessed and accepted by the Inspector. The test results shall be recorded and a copy shall be provided to Company.

5.5.1. Minimum Test Temperature

Hydrostatic test pressure shall be applied until the piping and its contents are at approximately the same temperature.

The minimum test temperature for hydrostatic pressure testing shall not be at metal temperatures below 2°C (35°F).

5.6. Test Records

Records shall be made of each piping system test. In general, these records shall include the following information:

- Date of test
- Identification of piping system
- Test medium
- Test pressure
- Minimum ambient temperature
- Certification by examiner

- Test duration
- Pressure and temperature recording charts and logs
- Test results
- Name of test operator
- Minimum metal temperature (if applicable)
- Test gauge calibration date

Forms, when completed, shall be retained in the Construction Contractor's QC file as a permanent record. A copy of each piping pressure test system record form shall be given to Company Authorized Inspector.

5.7. Test Completion

If leaks are found, their location shall be marked, the pressure shall be gradually released, and the piping shall be drained or vented. Appropriate repair or replacement shall be made. The affected piping shall be retested at the pressures originally specified for the test.

After completion of testing, all temporary blanks and blinds shall be removed, all operating blinds returned to proper position, and all lines and piping components (except those tested according to Section 5.4 of this Specification) shall be completely drained. Valves, valve internals, orifice plates, expansion joints, and short pieces of piping that have been removed shall be reinstalled with proper new gaskets in place. Valves that were closed during hydrotest shall be opened to ensure drainage of the bonnet cavity. Lines being drained after testing shall have all vents open. Piping systems downstream of check valves should be inspected to ensure complete drainage.

After lines have been drained, temporary supports shall be removed, and insulation and painting completed.

5.7.1. General

Instruments which were removed or blocked out for test shall be reinstalled and blocks placed in the normal operating position.

Temporary bolting and gaskets used for pressure testing shall be removed and replaced with line class bolts and gaskets.

Check valves that were jacked open or the internals were removed for pressure testing shall be returned to their proper operating position.

Instruments and process lead lines that were subjected to the hydrostatic pressure test shall be blown out with dry air or nitrogen.

Vent and drain connections that were added solely for pressure testing shall be closed and seal welded as required. Drains shall have either the valve closed and plugged or, if the valve is removed, the connection plugged and seal welded as required.

Painting and insulation shall be completed after inspection of seal welded vents and drains.

Strainer screens shall be removed, cleaned, and reinstalled.